Python — if / elif / else

Making Decisions in Code!

What Are Decisions?

Every day, you make decisions!

- If it rains → take an umbrella
- If it's sunny → wear sunglasses
- Otherwise → just go outside!

Computers can make decisions too!

The if Statement

The "if" keyword checks a condition.
If the condition is True, the code runs!

```
age = 10
if age >= 6:
    print("You can go to school!")
```

What is a Condition?

A condition is a question that is either True or False.

Examples:

- 5 > 3 → True
- 2 > 7 → False
- 10 == 10 → True

Comparison Operators

== equal to

!= not equal to

> greater than

< less than

>= greater than or equal

<= less than or equal

The else Statement

What if the condition is False?
Use "else" for the other option!

```
temperature = 15
if temperature > 30:
    print("It is hot!")
else:
    print("It is cool!")
```

The elif Statement

What if there are MORE than 2 choices?

Use "elif" (else if)!

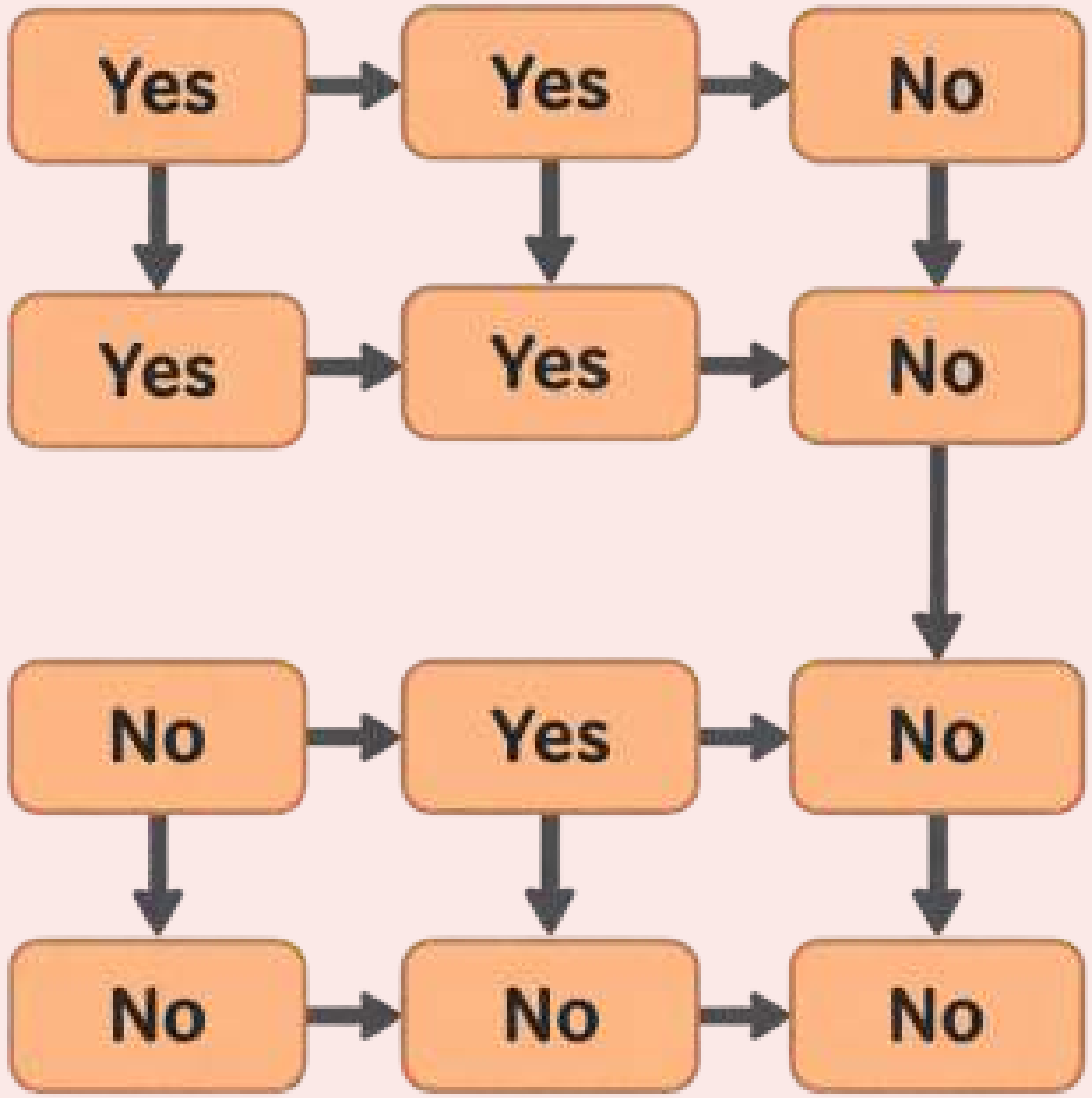
```
marks = 75
if marks >= 90:
    print("Excellent!")
elif marks >= 60:
    print("Good job!")
else:
    print("Keep trying!")
```

How It Works

Python checks conditions TOP to BOTTOM:

1. Check 'if' first
2. If False, check 'elif'
3. If all False, run 'else'

Only ONE block runs!



Indentation Matters!

The code INSIDE if/elif/else must be indented (4 spaces).

Correct:

```
if age > 5:
    print('Hello!')
```

Wrong:

```
if age > 5:
print('Hello!')  <- ERROR!
```

Let's Try: Fruit Shop

```
fruit = "mango"

if fruit == "mango":
    print("Yummy mango!")
elif fruit == "apple":
    print("Crunchy apple!")
else:
    print("Nice fruit!")
```

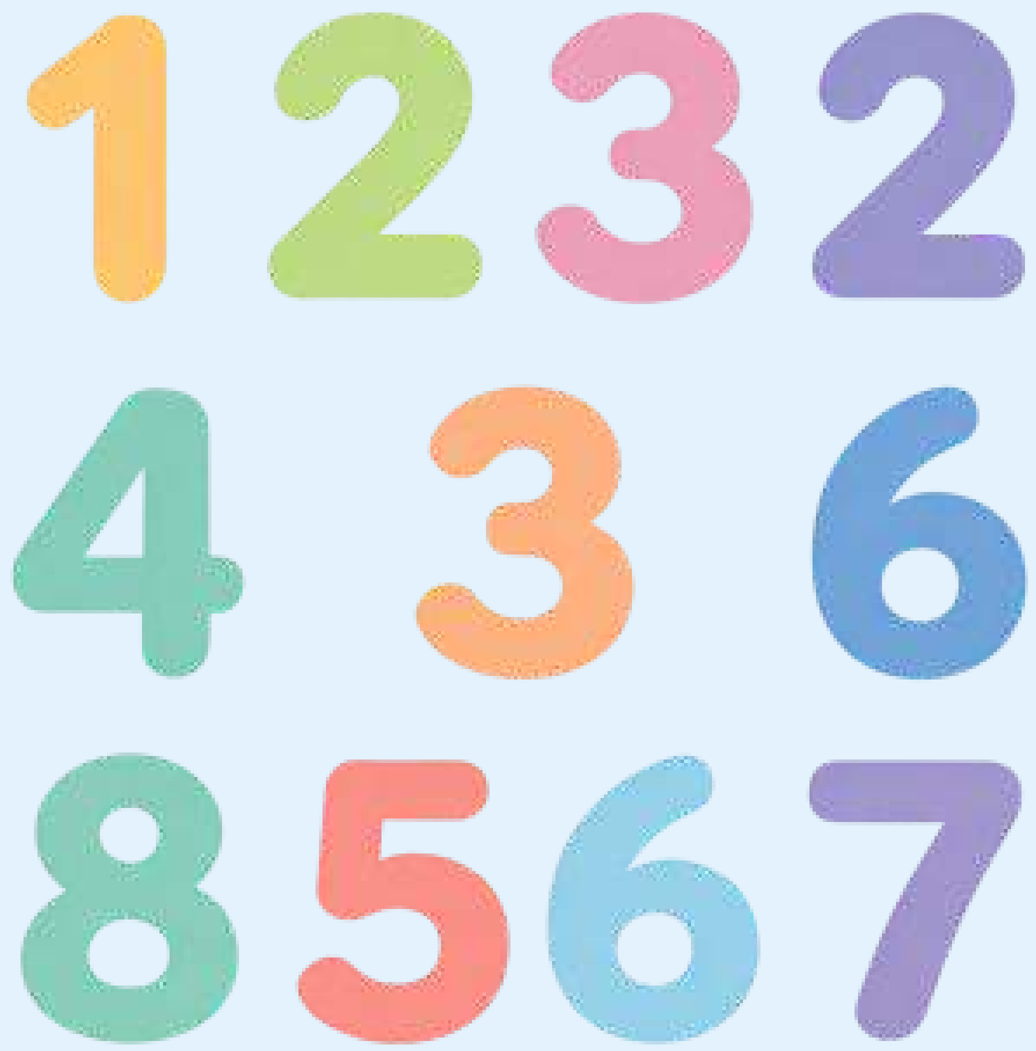
Let's Try: Even or Odd?

```
number = 7

if number % 2 == 0:
    print("Even number!")
else:
    print("Odd number!")
```

% gives the remainder.

7 / 2 = 3 remainder 1 -> Odd!



Multiple Conditions

You can use "and" & "or" to combine!

```
age = 10
has_ticket = True

if age >= 6 and has_ticket:
    print("Welcome to the show!")
else:
    print("Sorry, not today.")
```

Nested if Statements

You can put an "if" inside another "if"!

```
has_money = True
price = 20

if has_money:
    if price <= 50:
        print("You can buy it!")
    else:
        print("Too expensive!")
```

Common Mistakes

Forgetting the colon :

```
if age > 5     <- missing :
```

Wrong indentation

Using = instead of ==

```
if x = 5  <- wrong!
if x == 5  <- correct!
```

Quick Quiz!

What will this print?

```
x = 15
if x > 20:
    print('Big')
elif x > 10:
    print('Medium')
else:
    print('Small')
```

Answer: Medium!

Great Job!

Now you can make Python think and decide!

Practice Challenge:
Write a program that checks if a number is positive, negative, or zero.